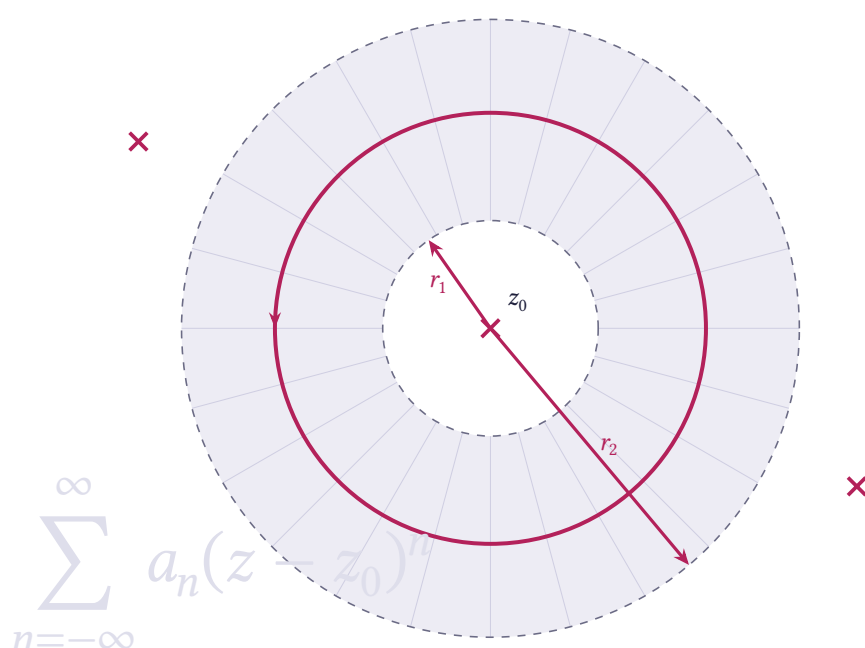


Laurent Series, Singularities and Residues

سری لورنت، نقاط تکین و مانده‌ها



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6.1 The Laurent Series

Theorem 6.1 ■ Laurent series

If the function $f(z)$ is analytic in the domain D , its Laurent series about the point z_0 is written

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{n+1}} dv$$

where C is a circle centred at z_0 and the function is analytic at all points inside and on this circle, except possibly at z_0 .

The radius of convergence is the distance from z_0 to the nearest non-analytic point of $f(z)$.

For non-negative n we have

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{n+1}} dv = \frac{f^{(n)}(z_0)}{n!}, \quad n = 0, 1, 2, \dots$$

Therefore

$$\begin{aligned} f(z) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=-\infty}^{-1} a_n (z - z_0)^n \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n} \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=1}^{\infty} \frac{A_n}{(z - z_0)^n}, \quad A_n = a_{-n} \end{aligned}$$

$$A_n = a_{-n} = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{-n+1}} dv = \frac{1}{2\pi i} \oint_C (v - z_0)^{n-1} f(v) dv, \quad n = 1, 2, \dots$$

Key result 6.2 ■ The Laurent series: Taylor part and principal part

$$f(z) = \underbrace{\sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n}_{\text{Taylor series part}} + \underbrace{\sum_{n=1}^{\infty} \frac{A_n}{(z - z_0)^n}}_{\text{principal part}}, \quad A_n = \frac{1}{2\pi i} \oint_C (v - z_0)^{n-1} f(v) dv$$

If z_0 is an irregular (singular) point, in other words if the function is not analytic there, the Taylor part alone is not sufficient to describe the function.

■ The Different Cases for the Point z_0

1. If all the A_n are zero, the point z_0 is called a **regular** point. In this case the Laurent series is just the Taylor series.
2. If $A_p \neq 0$ but $(A_n = 0 \ \forall n > p)$, then z_0 is called a **pole of order p** .
3. If z_0 is not a regular point and is also not a pole (in other words, no p can be found such that $A_p \neq 0$ and $(A_n = 0 \ \forall n > p)$), then z_0 is called an **essential singularity**.

6.2 The Residue

Definition 6.3 ■ Residue

The residue of $f(z)$ at the point z_0 is A_1 .

$$\text{Res}\{f(z)\}\Big|_{z=z_0} = A_1 = \frac{1}{2\pi i} \oint_C f(v) dv$$

Theorem 6.4

If the function $f(z)$ has a pole of order p at z_0 , then

$$A_k = \frac{1}{(p-k)!} \frac{d^{p-k}}{dz^{p-k}} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0}, \quad k = 1, 2, \dots, p$$

Proof.

$$f(z) = \sum_{n=1}^{\infty} \frac{A_n}{(z-z_0)^n} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n$$

If $f(z)$ has a pole of order p at z_0 we have

$$f(z) = \frac{A_p}{(z-z_0)^p} + \frac{A_{p-1}}{(z-z_0)^{p-1}} + \dots + \frac{A_2}{(z-z_0)^2} + \frac{A_1}{z-z_0} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n$$

Multiplying both sides by $(z-z_0)^p$ gives

$$(z-z_0)^p f(z) = A_p + A_{p-1}(z-z_0) + \dots + A_2(z-z_0)^{p-2} + A_1(z-z_0)^{p-1} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p}$$

Now if we set $z = z_0$ we get

$$(z-z_0)^p f(z) \Big|_{z=z_0} = A_p$$

If we differentiate $(z-z_0)^p f(z)$ once and then set $z = z_0$:

$$\begin{aligned} \frac{d}{dz} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} &= \left\{ A_{p-1} + \dots + A_2(p-2)(z-z_0)^{p-3} + A_1(p-1)(z-z_0)^{p-2} \right. \\ &\quad \left. + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p-1} (n+p) \right\} \Big|_{z=z_0} = A_{p-1} \end{aligned}$$

If we differentiate $(z-z_0)^p f(z)$ twice and then set $z = z_0$:

$$\begin{aligned} \frac{d^2}{dz^2} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} &= \left\{ 2A_{p-2} + \dots + A_2(p-2)(p-3)(z-z_0)^{p-4} + A_1(p-1)(p-2)(z-z_0)^{p-3} \right. \\ &\quad \left. + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p-2} (n+p)(n+p-1) \right\} \Big|_{z=z_0} = 2A_{p-2} \\ \implies A_{p-2} &= \frac{1}{2!} \frac{d^2}{dz^2} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} \end{aligned}$$

Exercise 6.5

Continue the differentiation in the same way and complete the proof of the theorem.

Consequence

In particular, when $f(z)$ has a pole of order p at z_0 , the residue is given by

$$\text{Res}\{f(z)\} = A_1 = \frac{1}{(p-1)!} \frac{d^{p-1}}{dz^{p-1}} [(z-z_0)^p f(z)] \Big|_{z=z_0}$$

6.3 The Residue Theorem**Theorem 6.6 ■ The residue theorem**

If the function $f(z)$ is analytic everywhere inside the path C except at the points $z_1, z_2, \dots, z_n$, and the residues of the function at these points are $A_1^1, A_1^2, \dots, A_1^n$ respectively, then

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n A_1^k$$

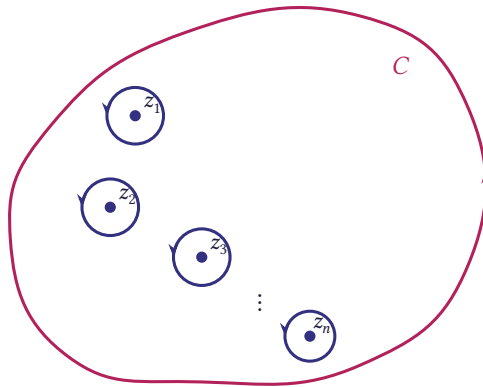


Figure 6.1 A path C and the singular points $z_1, \dots, z_n$ inside it; a small circle is drawn around each point

The proof is left to the reader as an exercise.

6.4 Examples of Taylor and Laurent Expansions

Example 6.7

Find the Laurent series of $f(z) = \frac{1}{z^4 + 1}$ in a neighbourhood of $z_0 = 0$.

We see that $f(z)$ is analytic at $z_0 = 0$, so this is a regular point and the Laurent series is the Taylor (Maclaurin) series. Hence

$$f(z) = f(0) + \frac{1}{1!}f^{(1)}(0)z + \frac{1}{2!}f^{(2)}(0)z^2 + \frac{1}{3!}f^{(3)}(0)z^3 + \dots$$

Geometric series:

$$1 + q + q^2 + q^3 + q^4 + \dots = \frac{1}{1 - q}, \quad |q| < 1$$

$$\frac{1}{1 + z^4} = 1 - z^4 + z^8 - z^{12} + \dots, \quad |z^4| < 1 \implies |z| < 1$$

(Taylor/Maclaurin series; region of convergence of radius 1.)

$$z^4 + 1 = 0 \implies z_1 = e^{i\pi/4}, \quad z_2 = e^{i3\pi/4}, \quad z_3 = e^{i5\pi/4}, \quad z_4 = e^{i7\pi/4}$$

We see that the distance from $z_0 = 0$ to the nearest non-analytic point of the function (the radius of convergence) is 1.

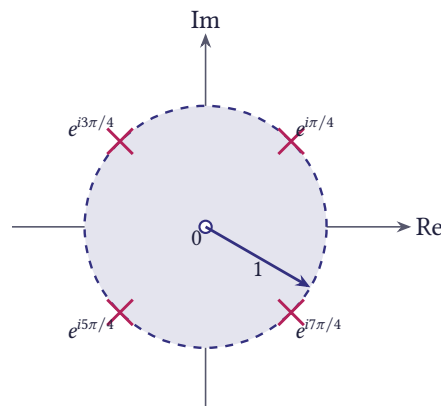


Figure 6.2 The four poles of $z^4 + 1 = 0$ on the unit circle; the radius of convergence of the series about the origin is 1

Example 6.8

Find the Laurent series of $f(z) = \frac{4z^2 + 30z + 68}{(z+4)^2(z-2)}$ in a neighbourhood of $z_0 = 0$.

We see that $f(z)$ is analytic at $z_0 = 0$, so this is a regular point and the Laurent series is the Taylor (Maclaurin) series. Hence

$$f(z) = f(0) + \frac{1}{1!}f^{(1)}(0)z + \frac{1}{2!}f^{(2)}(0)z^2 + \frac{1}{3!}f^{(3)}(0)z^3 + \dots$$

$$\frac{1}{z-2} = \frac{-1}{2-z} = \frac{-1}{2} \frac{1}{1-z/2} = \frac{-1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right), \quad \left| \frac{z}{2} \right| < 1 \implies |z| < 2$$

$$\frac{1}{4+z} = \frac{1}{4} \frac{1}{1+z/4} = \frac{1}{4} \left(1 - \frac{z}{4} + \frac{z^2}{16} - \frac{z^3}{64} + \dots \right), \quad \left| -\frac{z}{4} \right| < 1 \implies |z| < 4$$

$$\frac{d}{dz} \left(\frac{1}{4+z} \right) = \frac{-1}{(4+z)^2} = \frac{1}{4} \left(-\frac{1}{4} + \frac{2z}{16} - \frac{3z^2}{64} + \dots \right) \implies \frac{1}{(4+z)^2} = \frac{1}{4} \left(\frac{1}{4} - \frac{2z}{16} + \frac{3z^2}{64} - \dots \right)$$

$$f(z) = (4z^2 + 30z + 68) \times \frac{1}{(z+4)^2} \times \frac{1}{z-2}$$

$$f(z) = (4z^2 + 30z + 68) \times \left(\frac{-1}{2} \right) \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right) \times \frac{1}{4} \left(\frac{1}{4} - \frac{z}{8} + \frac{3z^2}{64} - \dots \right)$$

$$f(z) = \frac{-1}{16} \left(34 + 15z + \frac{67}{8}z^2 + \dots \right), \quad |z| < 2$$

(region of convergence of radius 2.)

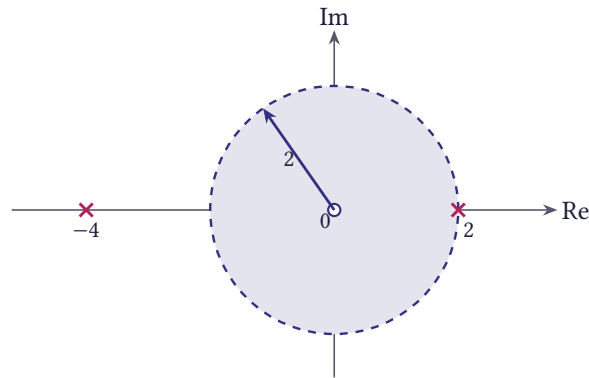


Figure 6.3 The poles $z = -4$ and $z = 2$; the circle of convergence about the origin has radius 2

Example 6.9

Find the Laurent series of $f(z) = \cos z^2$ in a neighbourhood of $z_0 = 0$.

From the definition of the cosine function,

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots$$

$$f(z) = \cos z^2 = 1 - \frac{z^4}{2!} + \frac{z^8}{4!} - \frac{z^{12}}{6!} + \dots$$

The region of convergence is the whole complex plane, with infinite radius of convergence.

Example 6.10

Find the Laurent series of $f(z) = \frac{\cos z}{1 - z^2}$ in a neighbourhood of $z_0 = 0$.

$$\begin{cases} \frac{1}{1 - z^2} = 1 + z^2 + z^4 + z^6 + \dots & |z^2| < 1 \quad |z| < 1 \\ \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots & |z| < \infty \end{cases}$$

$$\begin{aligned} f(z) = \frac{\cos z}{1 - z^2} &= (1 + z^2 + z^4 + z^6 + \dots) \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots \right) \\ &= 1 + \left(1 - \frac{1}{2!} \right) z^2 + \left(1 + \frac{1}{4!} - \frac{1}{2!} \right) z^4 + \dots, \quad |z| < 1 \end{aligned}$$

Example 6.11

Find the Laurent series of $f(z) = \frac{1}{z^4}$ in a neighbourhood of $z_0 = 1$.

$$g(z) = \frac{1}{z} = \frac{1}{z - 1 + 1} = \frac{1}{1 + (z - 1)} = 1 - (z - 1) + (z - 1)^2 - (z - 1)^3 + \dots, \quad |z - 1| < 1$$

$$\frac{d}{dz} g(z) = \frac{-1}{z^2} = -1 + 2(z - 1) - 3(z - 1)^2 + 4(z - 1)^3 - \dots, \quad |z - 1| < 1$$

$$\frac{d^2}{dz^2} g(z) = \frac{2}{z^3} = 2 - 6(z - 1) + 12(z - 1)^2 - 20(z - 1)^3 + \dots, \quad |z - 1| < 1$$

$$\frac{d^3}{dz^3} g(z) = \frac{-6}{z^4} = -6 + 24(z - 1) - 60(z - 1)^2 + \dots, \quad |z - 1| < 1$$

$$f(z) = \frac{1}{z^4} = 1 - 4(z - 1) + 10(z - 1)^2 - \dots$$

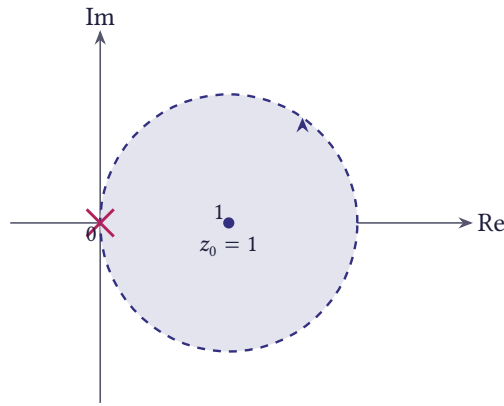


Figure 6.4 The region of convergence of the series about $z_0 = 1$ is a disc of radius 1 (up to the pole $z = 0$)

Exercise 6.12

Find the Laurent expansions of the following functions in a neighbourhood of $z_0 = 0$:

1. $f_1(z) = \frac{1}{\sqrt{1-z^2}}$
2. $f_2(z) = \sin^{-1} z$
3. $f_3(z) = \frac{e^z}{z^2(z^2+1)}$
4. Expand $\sinh z$ in powers of $(z - \pi i)$ and prove that $\lim_{z \rightarrow \pi i} \frac{\sinh z}{z - \pi i} = -1$
5. Prove that $z \cosh z^2 = z + \sum_{n=1}^{\infty} \frac{1}{(2n)!} z^{4n+1}$

6.5 Examples of Singular Points and Residues

Example 6.13

Find the Laurent series of $f(z) = \frac{z}{(z+1)(z+4)^3}$ in the neighbourhoods of its non-analytic points, and find the residue at those points.

The function is not analytic at $z = -1$ and $z = -4$.

(a) In a neighbourhood of $z = -1$, a circle centred at this point with radius less than 3 ($|z + 1| < 3$) does not contain the isolated singular point $z = -4$.

$$\begin{aligned} \frac{z}{(z+4)^3} &= \frac{z+4-4}{(z+4)^3} = \frac{z+4}{(z+4)^3} + \frac{-4}{(z+4)^3} = \frac{1}{(z+4)^2} - \frac{4}{(z+4)^3} \\ \frac{1}{z+4} &= \frac{1}{(z+1)+3} = \frac{1}{3} \frac{1}{1+\frac{z+1}{3}} = \frac{1}{3} \left[1 - \frac{z+1}{3} + \left(\frac{z+1}{3}\right)^2 - \left(\frac{z+1}{3}\right)^3 + \dots \right] \end{aligned}$$

$$\begin{aligned}
\frac{-1}{(z+4)^2} &= \frac{1}{3} \left[-\frac{1}{3} + \frac{2}{3^2}(z+1) - \frac{3}{3^3}(z+1)^2 + \dots \right] \Rightarrow \frac{1}{(z+4)^2} \\
&= \frac{1}{9} - \frac{2}{27}(z+1) + \frac{1}{27}(z+1)^2 - \dots \\
\frac{1}{(z+4)^3} &= \frac{1}{6} \left[\frac{2}{3^2} - \frac{2 \times 3}{3^3}(z+1) + \frac{3 \times 4}{3^4}(z+1)^2 - \frac{4 \times 5}{3^5}(z+1)^3 + \dots \right] \\
&= \frac{1}{27} - \frac{1}{27}(z+1) + \frac{2}{81}(z+1)^2 - \dots \\
\frac{z}{(z+4)^3} &= \frac{1}{(z+4)^2} - \frac{4}{(z+4)^3} = -\frac{1}{27} + \frac{2}{27}(z+1) - \frac{5}{81}(z+1)^2 + \dots \\
f(z) &= \frac{1}{z+1} \frac{z}{(z+4)^3} = -\frac{1}{27} \frac{1}{z+1} + \frac{2}{27} - \frac{5}{81}(z+1) + \dots
\end{aligned}$$

Type of singular point: a simple pole (of order 1). Residue: $\text{Res}\{f(z)\}_{z=-1} = -\frac{1}{27}$.

(b) In a neighbourhood of $z = -4$, a circle centred at this point with radius less than 3 ($|z+4| < 3$) does not contain the isolated singular point $z = -1$.

$$\begin{aligned}
\frac{z}{z+1} &= \frac{z+1-1}{z+1} = \frac{z+1}{z+1} - \frac{1}{z+1} = 1 - \frac{1}{z+1} \\
\frac{1}{z+1} &= \frac{1}{z+4-3} = \frac{1}{-3+(z+4)} = \frac{-1}{3} \frac{1}{1-\frac{z+4}{3}} = \frac{-1}{3} \left[1 + \frac{z+4}{3} + \frac{(z+4)^2}{3^2} + \dots \right], \\
|z+4| &< 3
\end{aligned}$$

$$\begin{aligned}
\frac{z}{z+1} &= 1 + \frac{1}{3} + \frac{1}{3^2}(z+4) + \frac{1}{3^3}(z+4)^2 + \frac{1}{3^4}(z+4)^3 + \dots \\
f(z) &= \frac{z}{(z+1)(z+4)^3} = \frac{1}{(z+4)^3} \frac{z}{z+1} \\
&= \frac{4/3}{(z+4)^3} + \frac{1}{3^2} \frac{1}{(z+4)^2} + \frac{1}{3^3} \frac{1}{z+4} + \frac{1}{3^4} + \frac{1}{3^5}(z+4) + \frac{1}{3^6}(z+4)^2 + \dots
\end{aligned}$$

Type of singular point: a pole of order 3. Residue: $\text{Res}\{f(z)\}_{z=-4} = \frac{1}{27}$.

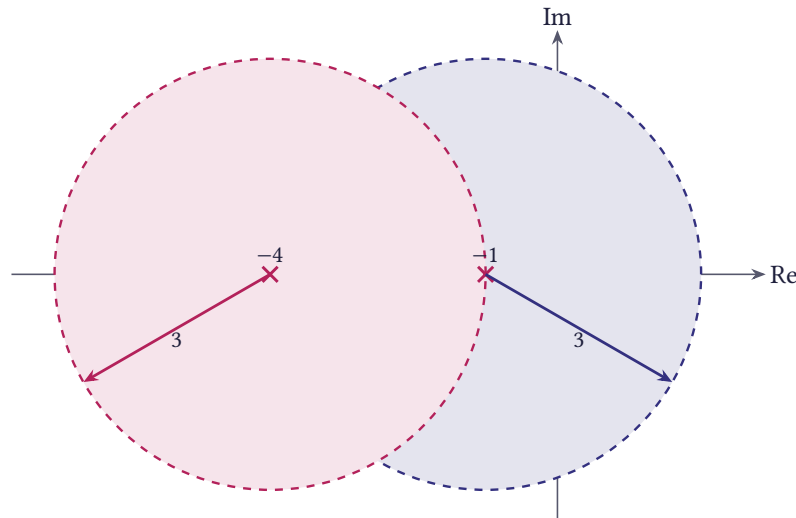


Figure 6.5 The poles $z = -1$ and $z = -4$ of $f(z) = \frac{z}{(z+1)(z+4)^3}$

Example 6.14

Find the Laurent series of $f(z) = e^{1/z}$ in a neighbourhood of its non-analytic point, and find the residue at that point.

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \implies e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots$$

Type of singular point: an essential singularity (infinitely many negative powers). Residue: $\text{Res}\{f(z)\}_{z=0} = 1$.

Example 6.15

Find the Laurent series of $f(z) = \cos\left(\frac{z}{z-1}\right)$ in a neighbourhood of its non-analytic point, and find the residue at that point.

$$\frac{z}{z-1} = 1 + \frac{1}{z-1} \implies \cos\left(\frac{z}{z-1}\right) = \cos\left(1 + \frac{1}{z-1}\right) = \cos 1 \cos\left(\frac{1}{z-1}\right) - \sin 1 \sin\left(\frac{1}{z-1}\right)$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots, \quad \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

$$\begin{aligned} \cos\left(\frac{z}{z-1}\right) &= \cos 1 \left(1 - \frac{1}{2!} \frac{1}{(z-1)^2} + \frac{1}{4!} \frac{1}{(z-1)^4} - \dots\right) \\ &\quad - \sin 1 \left(\frac{1}{z-1} - \frac{1}{3!} \frac{1}{(z-1)^3} + \frac{1}{5!} \frac{1}{(z-1)^5} - \dots\right) \\ &= \cos 1 - \sin 1 \frac{1}{z-1} - \cos 1 \frac{1}{2!} \frac{1}{(z-1)^2} + \sin 1 \frac{1}{3!} \frac{1}{(z-1)^3} + \dots \end{aligned}$$

Type of singular point: an essential singularity. Residue: $\text{Res}\{f(z)\}_{z=1} = -\sin 1$.

Theorem 6.16 ■ Residue at a simple pole

Prove that if $f(z) = q(z)/g(z)$ and $g(z)$ has a simple root at z_0 , then

$$\text{Res}\{f(z)\}\Big|_{z=z_0} = \frac{q(z_0)}{g'(z_0)}$$

Proof.

Since z_0 is a pole of order 1 we have ($p = 1$):

$$\begin{aligned} \text{Res}\{f(z)\}\Big|_{z=z_0} &= A_1 = \frac{1}{(p-1)!} \frac{d^{p-1}}{dz^{p-1}} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} = (z-z_0) \frac{q(z)}{g(z)} \Big|_{z=z_0} \\ &= (z-z_0) \frac{q(z)}{g(z) - g(z_0)} \Big|_{z=z_0} = \frac{q(z)}{\frac{g(z) - g(z_0)}{z-z_0}} \Big|_{z=z_0} = \frac{q(z_0)}{g'(z_0)} \end{aligned}$$

■

Exercise 6.17

Find the Laurent series of the following functions about $z = 0$ and determine the type of this point.

$$f_1(z) = \frac{z}{e^z - 1}, \quad f_2(z) = \frac{e^{-z}}{z^3}, \quad f_3(z) = \frac{1}{z^2(1 - z^2)}, \quad f_4(z) = \frac{\sin z^2}{z^3}$$